

16/10/2025

Procédure

pfSense

Table des matières

Relier le pfSense à l'AD	3
Configuration d'OpenVPN.....	9
Gestion des certificats	9
Création des utilisateurs locaux.....	11
Configurer le serveur OpenVPN	12
Exporter la configuration OpenVPN	15
Créer les règles de firewall pour OpenVPN	16
Autoriser le flux OpenVPN.....	16
Autoriser les flux vers les ressources	17
Test de l'accès en VPN depuis le poste	18

Valentin TRILLAUD

HPK

Vérifier que les 2 cartes réseaux soient sur le bon vlan :

Machine virtuelle 211003 (TRILLAUD-HPK-PFSENSE-003) sur le nœud pve13 Aucune étiquette

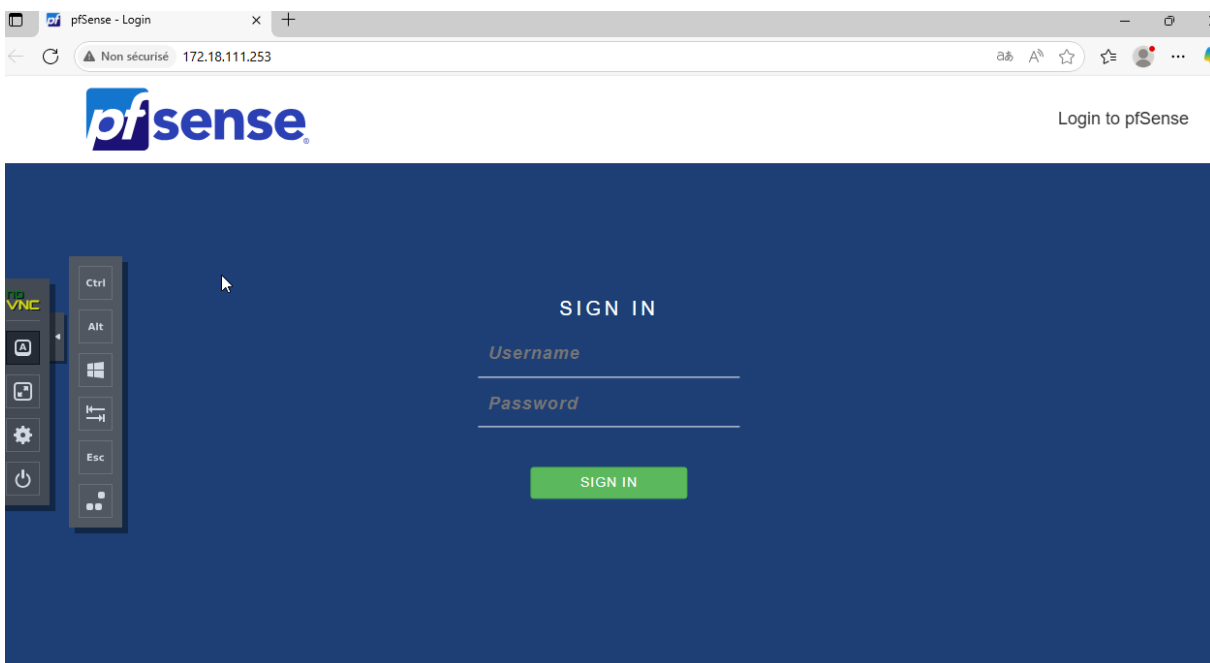
Résumé	Ajouter	Supprimer	Éditer	Action disque	Revenir en arrière
Console	Mémoire	2.00 Gio			
Matériel	Processeurs	2 (1 sockets, 2 cores) [x86-64-v2-AES]			
Cloud-Init	BIOS	Par défaut (SeaBIOS)			
Options	Affichage	Par défaut			
Historique des tâches	Machine	Par défaut (i440fx)			
Instantanés	Contrôleur SCSI	VirtIO SCSI single			
Pare-feu	Disque dur (virtio0)	DATASTORE-PVE13:vm-211003-disk-0,discard=on,iothread=1,size=32G			
	Carte réseau (net0)	e1000=BC:24:11:11:97:93,bridge=vlan111,firewall=1			
	Carte réseau (net1)	e1000=BC:24:11:38:03:F1,bridge=vlan211,firewall=1			

Configurer les 2 cartes réseaux sur pfSense :

```
* Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***
WAN (wan) -> em0 -> v4/DHCP4: 172.20.111.1/24
LAN (lan) -> em1 -> v4: 172.18.111.253/24
```

Le résultat du ping ne s'affiche pas sur pfsense, il faut faire le ping depuis le routeur ou le switch.

Se connecter à l'interface du pfSense via un moteur de recherche :



Login : admin Mot de passe : pfsense

Ajout d'une access-list ;

```
Router#sh access-list 10
Standard IP access list 10
 10 permit 192.168.111.0, wildcard bits 0.0.0.255
 20 permit 172.16.111.0, wildcard bits 0.0.0.255 (8512 matches)
Router#
```

Créer une règle pour le bureau à distance :

pfSense.home.arpa - Firewall: NAT

Non sécurisé 172.18.111.253/firewall_nat_edit.php?id=0

The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.

Firewall / NAT / Port Forward / Edit

Edit Redirect Entry

Disabled ☐ Disable this rule

No RDR (NOT) ☐ Disable redirection for traffic matching this rule
This option is rarely needed. Don't use this without thorough knowledge of the implications.

Interface WAN
Choose which interface this rule applies to. In most cases "WAN" is specified.

Address Family IPv4
Select the Internet Protocol version this rule applies to.

Protocol TCP
Choose which protocol this rule should match. In most cases "TCP" is specified.

Source [Display Advanced](#)

Destination ☐ Invert match. WAN address
Type Address/mask

Destination port range MS RDP
From port Custom To port Custom

pfSense.home.arpa - Firewall: NAT

Non sécurisé 172.18.111.253/firewall_nat_edit.php?id=0

Source port range Any
From port Custom To port Custom

Specify the source port or port range for this rule. This is usually random and almost never equal to the destination port range (and should usually be 'any'). The 'to' field may be left empty if only filtering a single port.

Destination ☐ Invert match. WAN address
Type Address/mask

Destination port range MS RDP
From port Custom To port Custom

Specify the port or port range for the destination of the packet for this mapping. The 'to' field may be left empty if only mapping a single port.

Redirect target IP Address or Alias
Type Address
172.16.111.1

Enter the internal IP address of the server on which to map the ports. e.g.: 192.168.1.12 for IPv4
In case of IPv6 addresses, it must be from the same "scope".
i.e. it is not possible to redirect from link-local addresses scope (fe80::) to local scope (::1)

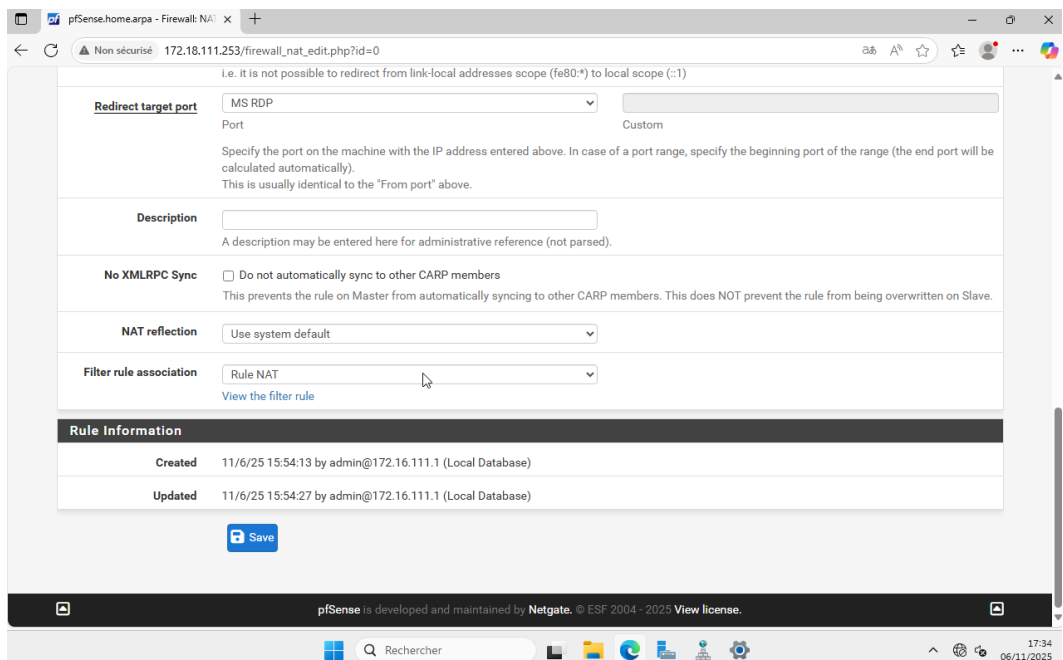
Redirect target port MS RDP
Port Custom

Specify the port on the machine with the IP address entered above. In case of a port range, specify the beginning port of the range (the end port will be calculated automatically).
This is usually identical to the "From port" above.

Description

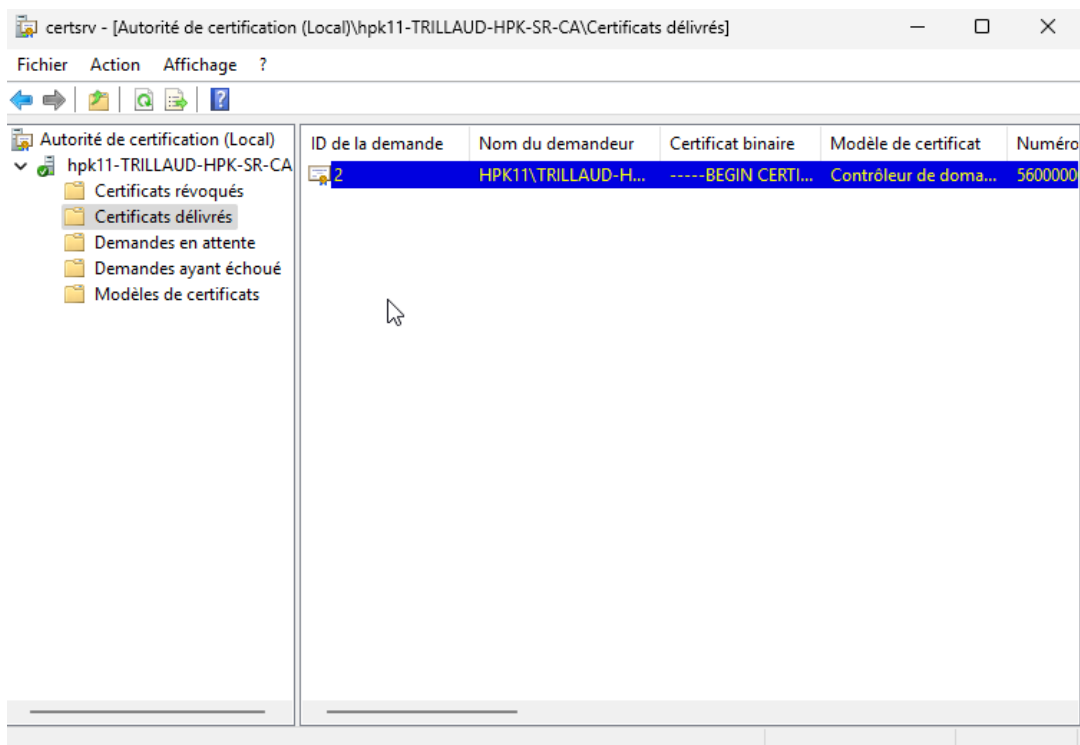
A description may be entered here for administrative reference (not parsed).

No XMLRPC Sync ☐ Do not automatically sync to other CARP members
This prevents the rule on Master from automatically syncing to other CARP members. This does NOT prevent the rule from being overwritten on Slave.



Relier le pfSense à l'AD

Ajouter le rôle d'Autorité de Certification :



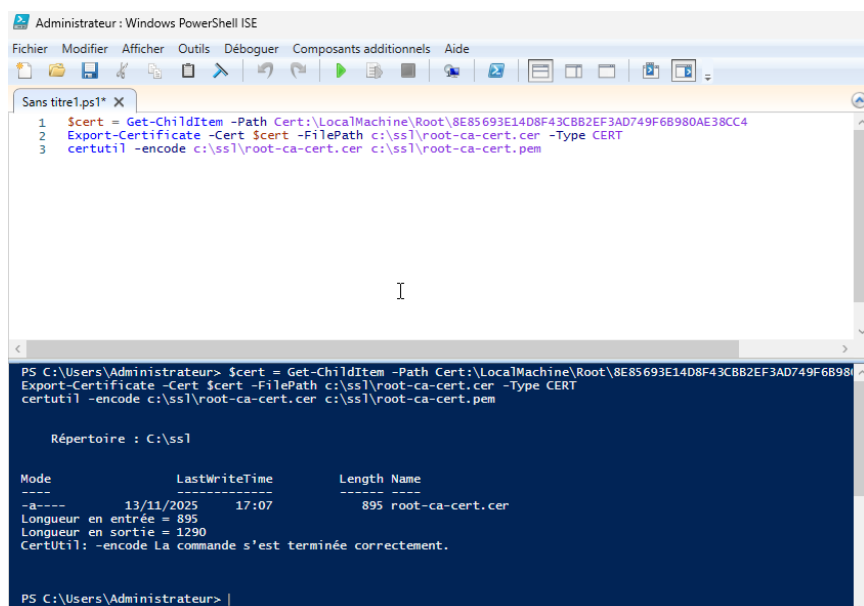
Afficher les certificats dans PowerShell :

```
PS C:\WINDOWS\system32> Get-ChildItem -Path Cert:\LocalMachine\Root

PSParentPath : Microsoft.PowerShell.Security\Certificate::LocalMachine\Root

Thumbprint                               Subject
-----
CDD4EEAE600AC7F40C3802C171E30148030C072 CN=Microsoft Root Certificate Authority, DC=microsoft, DC=com
BE36A4562F82EE05DB83D32323ADF445084ED656 CN=Thawte Timestamping CA, OU=Thawte Certification, O=Thawte, L=Durbanvill...
A43489159A520F0D93D032CCAF37E7FE20A8B419 CN=Microsoft Root Authority, OU=Microsoft Corporation, OU=Copyright (c) 19...
92B46C76E13054E104F230517E6E504D43AB10B5 CN=Symantec Enterprise Mobile Root for Microsoft, O=Symantec Corporation, ...
8F43288AD272F3103B6F81428485EA3014C0BCEFE CN=Microsoft Root Certificate Authority 2011, O=Microsoft Corporation, L=R...
7F88CD7223F3C818C994614A89C99FA3B5247 CN=Microsoft Authenticode(tm) Root Authority, O=MSFT, C=US
3B1EFD3A66EA28B16697394703A72CA340A05BD05 CN=Microsoft Root Certificate Authority 2010, O=Microsoft Corporation, L=R...
31F9FC8BA38059868721EA7295C65B3444534274 CN=Microsoft ECC TS Root Certificate Authority 2018, O=Microsoft Corporati...
245C97DF7514E7CF2DF8BE72AE95789E04741E85 OU=Copyright (c) 1997 Microsoft Corp., OU=Microsoft Time Stamping Service ...
18F7C1FCC3090203FD5BAA2F861A754976C8DD25 OU="NO LIABILITY ACCEPTED, (c)97 VeriSign, Inc.", OU=VeriSign Time Stampin...
06F1AA330B927B753A40E68CDF22E34BCBEF3352 CN=Microsoft ECC Product Root Certificate Authority 2018, O=Microsoft Corp...
0119E81BE9A14CD8E22F40AC118C687ECBA3F4D8 CN=Microsoft Time Stamp Root Certificate Authority 2014, O=Microsoft Corpo...
8E85693E14D8F43CB82EF3AD749F6B980AE38CC4 CN=hpk11-TRILLAUD-HPK-SR-CA, DC=hpk11, DC=labo
DF3C24F98FD666761B268073FE06D1CC8D4F82A4 CN=DigiCert Global Root G2, OU=www.digicert.com, O=DigiCert Inc, C=US
A8985D3A65E5E5C4B2D7D6D40C6DD2FB19C5436 CN=DigiCert Global Root CA, OU=www.digicert.com, O=DigiCert Inc, C=US
742C3192E607E424E84549542BE1B8C53E6174E2 OU=Class 3 Public Primary Certification Authority, O="VeriSign, Inc.", C=US
3679CA35668772304D30A5F8B730FA77BB70D54 CN=VeriSign Universal Root Certification Authority, OU="(c) 2008 VeriSign,...
0563B8630D62D75AB8C8AB1E4BDFB5A899B24D43 CN=DigiCert Assured ID Root CA, OU=www.digicert.com, O=DigiCert Inc, C=US
8E85693E14D8F43CB82EF3AD749F6B980AE38CC4 CN=hpk11-TRILLAUD-HPK-SR-CA, DC=hpk11, DC=labo
```

Exporter le certificat de l'AD dans un dossier SSL préalablement créé :



```

1 $cert = Get-ChildItem -Path Cert:\LocalMachine\Root\8E85693E14D8F43CB82EF3AD749F6B980AE38CC4
2 Export-Certificate -Cert $cert -FilePath c:\ssl\root-ca-cert.cer -Type CERT
3 certutil -encode c:\ssl\root-ca-cert.cer c:\ssl\root-ca-cert.pem

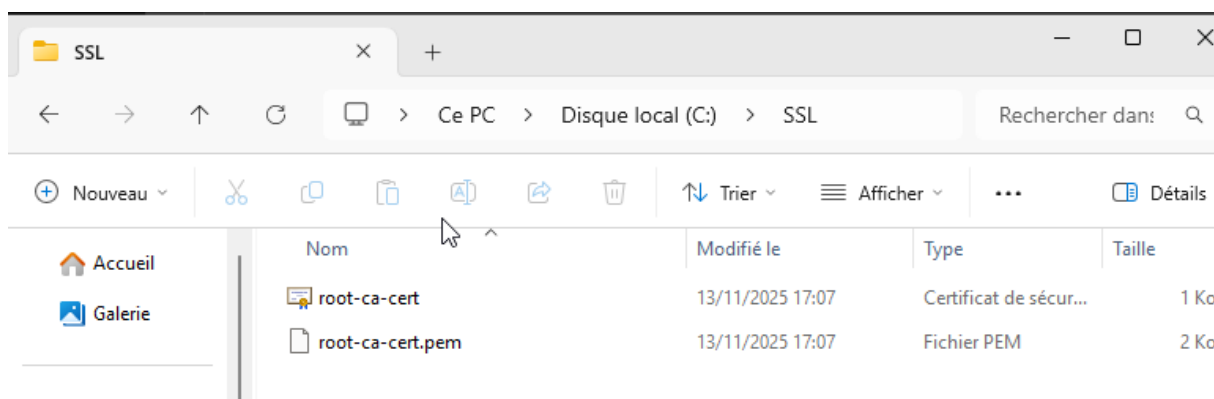
PS C:\Users\Administrateur> $cert = Get-ChildItem -Path Cert:\LocalMachine\Root\8E85693E14D8F43CB82EF3AD749F6B980AE38CC4
Export-Certificate -Cert $cert -FilePath c:\ssl\root-ca-cert.cer -Type CERT
certutil -encode c:\ssl\root-ca-cert.cer c:\ssl\root-ca-cert.pem

Répertoire : C:\ssl

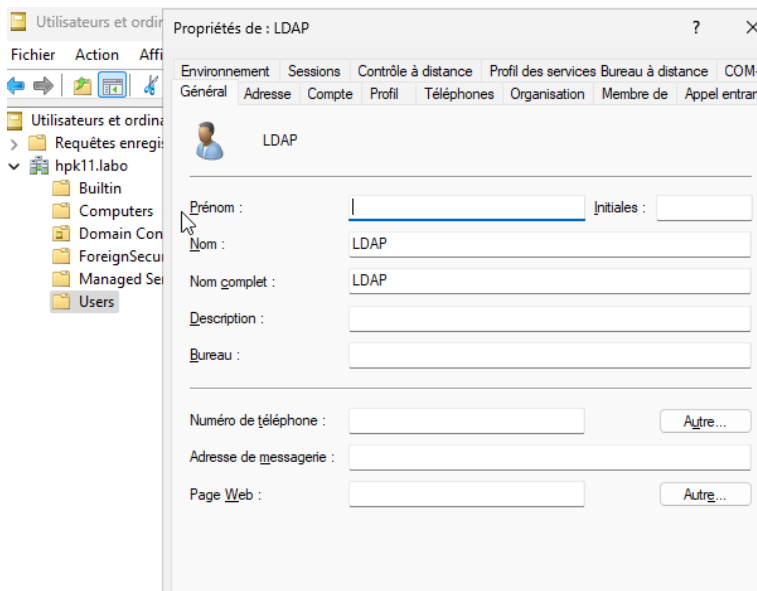
Mode                LastWriteTime         Length Name
----                -
-a-----      13/11/2025   17:07           895 root-ca-cert.cer
Longueur en entrée = 895
Longueur en sortie = 1290
CertUtil: -encode La commande s'est terminée correctement.

PS C:\Users\Administrateur>

```



Créer un utilisateur pour faire la liaison avec LDAP (mdp durée de vie indéfinie) :



Importer le certificat dans pfSense :

System / Certificates / Authorities / Edit

Authorities Certificates Revocation

Create / Edit CA

Descriptive name
 The name of this entry as displayed in the GUI for reference.
 This name can contain spaces but it cannot contain any of the following characters: ?, >, <, &, /, \, ", '.

Method

Trust Store ☐ Add this Certificate Authority to the Operating System Trust Store
 When enabled, the contents of the CA will be added to the trust store so that they will be trusted by the operating system.

Randomize Serial ☐ Use random serial numbers when signing certificates
 When enabled, if this CA is capable of signing certificates then serial numbers for certificates signed by this CA will be automatically randomized and checked for uniqueness instead of using the sequential value from Next Certificate Serial.

Existing Certificate Authority

Certificate data

Configurer le DNS :

System / General Setup

System

Hostname

pfSense

Name of the firewall host, without domain part.

Domain

home.arpa

Domain name for the firewall.

Do not end the domain name with '.local' as the final part (Top Level Domain, TLD). The 'local' TLD is widely used by mDNS (e.g. Avahi, Bonjour, Rendezvous, Airprint, Airplay) and some Windows systems and networked devices. These will not network correctly if the router uses 'local' as its TLD. Alternatives such as 'home.arpa', 'local.lan', or 'mylocal' are safe.

DNS Server Settings

DNS Servers

172.16.111.1

Address

Enter IP addresses to be used by the system for DNS resolution. These are also used for the DHCP service, DNS Forwarder and DNS Resolver when it has DNS Query Forwarding enabled.

TRILLAUD-HPK-SR.hpk1

Hostname

Enter the DNS Server Hostname for TLS Verification in the DNS Resolver (optional).

none

Gateway

Optionally select the gateway for each DNS server. When using multiple WAN connections there should be at least one unique DNS server per gateway.

Et ajouter un Host Overrides dans Services > DNS Resolver :

Save

Host Overrides

Host	Parent domain of host	IP to return for host	Description	Actions
TRILLAUD-HPK-SR	hpk11.labo	172.16.111.1		 

Enter any individual hosts for which the resolver's standard DNS lookup process should be overridden and a specific IPv4 or IPv6 address should automatically be returned by the resolver. Standard and also non-standard names and parent domains can be entered, such as 'test', 'nas.home.arpa', 'mycompany.localdomain', '1.168.192.in-addr.arpa', or 'somesite.com'. Any lookup attempt for the host will automatically return the given IP address, and the usual lookup server for the domain will not be queried for the host's records.

+ Add

Après ça, le ping du nom du serveur doit fonctionner.

Création du serveur d'authentification :

System / User Manager / Authentication Servers / Edit

Users Groups Settings Change Password Authentication Servers

Server Settings

Descriptive name

Liaison pfSense AD

Type

LDAP

LDAP Server Settings

Hostname or IP address

TRILLAUD-HPK-SRV-AD-DNS-001.hpk11.labo

NOTE: When using SSL/TLS or STARTTLS, this hostname MUST match a Subject Alternative Name (SAN) or the Common Name (CN) of the LDAP server SSL/TLS Certificate.

Port value

636

Transport

SSL/TLS Encrypted

Peer Certificate Authority

HPK-Root CA

This CA is used to validate the LDAP server certificate when 'SSL/TLS Encrypted' or 'STARTTLS Encrypted' Transport is active. This CA must match the CA used by the LDAP server.

This CA is used to validate the LDAP server certificate when 'SSL/TLS Encrypted' or 'STARTTLS Encrypted' Transport is active. This CA must match the CA used by the LDAP server.

Protocol version 3

Server Timeout 25
Timeout for LDAP operations (seconds)

Search scope Level
Entire Subtree

Base DN
OU=hpk,DC=hpk11,DC=labo

Authentication containers OU=securite,OU=hpk,DC=hpk11,DC=labo Q Select a container
Note: Semi-Colon separated. This will be prepended to the search base dn above or the full container path can be specified containing a dc= component.
Example: CN=Users;DC=example,DC=com or OU=Staff;OU=Freelancers

Extended query ☐ Enable extended query

Bind anonymous ☐ Use anonymous binds to resolve distinguished names

Bind credentials CN=pfsense,OU=securite,OU=hpk,DC=hpk11,DC=labo

User naming attribute samAccountName

Group naming attribute cn

Group naming attribute cn

Group member attribute memberOf

RFC 2307 Groups ☐ LDAP Server uses RFC 2307 style group membership
RFC 2307 style group membership has members listed on the group object rather than using groups listed on user object. Leave unchecked for Active Directory style group membership (RFC 2307bis).

Group Object Class posixGroup
Object class used for groups in RFC2307 mode. Typically "posixGroup" or "group".

Shell Authentication Group DN
If LDAP server is used for shell authentication, user must be a member of this group and have a valid posixAccount attributes to be able to login.
Example: CN=Remoteshellusers,CN=Users,DC=example,DC=com

UTF8 Encode ☐ UTF8 encode LDAP parameters before sending them to the server.
Required to support international characters, but may not be supported by every LDAP server.

Username Alterations ☐ Do not strip away parts of the username after the @ symbol
e.g. user@host becomes user when unchecked.

Allow unauthenticated bind ☐ Allow unauthenticated bind
Unauthenticated binds are bind with an existing login but with an empty password. Some LDAP servers (Microsoft AD) allow this type of bind without any possibility to disable it.

Save

Tester l'authentification avec l'utilisateur créé :

Diagnostics / Authentication 🔧 ?

User pfsense authenticated successfully. This user is a member of groups:

Authentication Test

Authentication Server Liaison pfSense AD
Select the authentication server to test against.

Username pfsense

Password

Debug ☐ Set debug flag
Sets the debug flag when performing authentication, which may trigger additional diagnostic entries in the system log (e.g. for LDAP).

Test

Tester l'authentification du serveur LADPS au PfSense dans System → User Manager → Settings :

LDAP settings



Test results

Attempting connection to	TRILLAUD-HPK-SRV-AD-DNS-001.hpk11.labo	OK
Attempting bind to	TRILLAUD-HPK-SRV-AD-DNS-001.hpk11.labo	OK
Attempting to fetch Organizational Units from	TRILLAUD-HPK-SRV-AD-DNS-001.hpk11.labo	OK
Organization units found OU=hpk,DC=hpk11,DC=labo OU=securite,OU=hpk,DC=hpk11,DC=labo		

Cr  er le m  me groupe dans PfSense et lui permettre d'afficher quelques options :

System / User Manager / Groups

Users Groups Settings Change Password Authentication Servers

Groups

Group name	Description	Member Count	Actions
Observateurs PfSense		0	
admins	System Administrators	1	
all	All Users	1	

+ Add

Group Properties

Group name

Observateurs PfSense

Scope

Remote

Warning: Changing this setting may affect the local groups file, in which case a reboot may be required for the changes to take effect.

Description

Group description, for administrative information only

Group membership

admin

Not members

Members

>> Move to "Members"

<< Move to "Not members"

Hold down CTRL (PC)/COMMAND (Mac) key to select multiple items.

Assigned Privileges

Name	Description	Action
WebCfg - Dashboard (all)	Allow access to all pages required for the dashboard.	
WebCfg - Status: DHCP leases	Allow access to the 'Status: DHCP leases' page.	

+ Add

Se connecter avec l'utilisateur de test pour v  rifier que cela marche :

SIGN IN

pfSense

SIGN IN

Status / Dashboard

System Information

Name	pfSense.home.arpa
User	pfSense@172.16.111.1 (LDAP/Liaison pfSense AD)
System	QEMU Guest Netgate Device ID: 4f83e9b660fbc67a3fb8
BIOS	Vendor: SeaBIOS Version: rel-1.16.3-0-ga6ed6b701f0a-prebuilt.qemu.org Release Date: Tue Apr 1 2014 Boot Method: BIOS
Version	2.8.1-RELEASE (amd64) built on Tue Sep 9 18:29:00 CEST 2025 FreeBSD 15.0-CURRENT The system is on the latest version. Version information updated at Thu Nov 27 16:56:41 CET 2025
CPU Type	QEMU Virtual CPU version 2.5+ 2 CPUs : 1 package(s) x 2 core(s) AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive

Netgate Services And Support

Contract type Community Support
Community Support Only

NETGATE AND pfSense COMMUNITY SUPPORT RESOURCES

If you purchased your pfSense gateway firewall appliance from Netgate and elected **Community Support** at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the [NETGATE RESOURCE LIBRARY](#).

You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.

- [Upgrade Your Support](#)
- [Community Support Resources](#)
- [Netgate Global Support FAQ](#)
- [Official pfSense Training by Netgate](#)
- [Netgate Professional Services](#)
- [Visit Netgate.com](#)

If you decide to purchase a Netgate Global TAC Support subscription, you **MUST** have your **Netgate Device ID (NDID)** from your firewall in order to.

Configuration d'OpenVPN

Gestion des certificats

Créer une autorité de certificat pour le vpn :

System / Certificates / Authorities

Authorities Certificates Revocation

Search

Search term Both

Enter a search string or *nix regular expression to search certificate names and distinguished names.

Certificate Authorities

Name	Internal	Issuer	Certificates	Distinguished Name	In Use	Actions
HPK-Root CA	✗	self-signed	0	DC=hpk11, DC=labo, CN=hpk11-TRILLAUD-HPK-SR-CA Valid From: Thu, 13 Nov 2025 16:48:50 +0100 Valid Until: Mon, 13 Nov 2045 16:58:50 +0100	LDAP Server	🔧

+ Add

Authorities

Certificates

Revocation

Create / Edit CA

Descriptive name

CA-HPK11-OPENVPN

The name of this entry as displayed in the GUI for reference.
This name can contain spaces but it cannot contain any of the following characters: ?, >, <, &, /, \, ", '.

Method

Create an internal Certificate Authority

Trust Store

☐ Add this Certificate Authority to the Operating System Trust Store
When enabled, the contents of the CA will be added to the trust store so that they will be trusted by the operating system.

Randomize Serial

☐ Use random serial numbers when signing certificates
When enabled, if this CA is capable of signing certificates then serial numbers for certificates signed by this CA will be automatically randomized and checked for uniqueness instead of using the sequential value from Next Certificate Serial.

System / Certificates / Authorities

Authorities

Certificates

Revocation

Search

Search term

Both

Search

Clear

Enter a search string or *nix regular expression to search certificate names and distinguished names.

Certificate Authorities

Name	Internal	Issuer	Certificates	Distinguished Name	In Use	Actions
HPK-Root CA	✗	self-signed	0	DC=hpk11, DC=labo, CN=hpk11-TRILLAUD-HPK-SR-CA ⓘ Valid From: Thu, 13 Nov 2025 16:48:50 +0100 Valid Until: Mon, 13 Nov 2045 16:58:50 +0100	LDAP Server	✎ ⚙
CA-HPK11-OPENVPN	✓	self-signed	0	CN=hpk11-OpenVPN, C=FR ⓘ Valid From: Thu, 27 Nov 2025 17:14:37 +0100 Valid Until: Sun, 25 Nov 2035 17:14:37 +0100		✎ ⚙ 🔑 🗑

Authorities

Certificates

Revocation

Add/Sign a New Certificate

Method

Create an internal Certificate

Descriptive name

VPN-SSL-REMOTE-ACCESS

The name of this entry as displayed in the GUI for reference.
This name can contain spaces but it cannot contain any of the following characters: '?', '>', '<', '&', '/', '\', ' '.

Internal Certificate

Certificate authority

CA-HPK11OPENVPN

Key type

RSA

2048

The length to use when generating a new RSA key, in bits.
The Key Length should not be lower than 2048 or some platforms may consider the certificate invalid.

Digest Algorithm	sha256	The digest method used when the certificate is signed. The best practice is to use SHA256 or higher. Some services and platforms, such as the GUI web server and OpenVPN, consider weaker digest algorithms invalid.
Lifetime (days)	3650	The length of time the signed certificate will be valid, in days. Server certificates should not have a lifetime over 398 days or some platforms may consider the certificate invalid.
Common Name	hpk11-serveur-vpn	
The following certificate subject components are optional and may be left blank.		
Country Code	FR	
State or Province	e.g. Texas	
City	e.g. Austin	
Organization	e.g. My Company Inc	
Organizational Unit	e.g. My Department Name (optional)	

Certificate Attributes	
Attribute Notes	The following attributes are added to certificates and requests when they are created or signed. These attributes behave differently depending on the selected mode. For Internal Certificates, these attributes are added directly to the certificate as shown.
Certificate Type	Server Certificate Add type-specific usage attributes to the signed certificate. Used for placing usage restrictions on, or granting abilities to, the signed certificate.
Alternative Names	FQDN or Hostname Type Value Enter additional identifiers for the certificate in this list. The Common Name field is automatically added to the certificate as an Alternative Name. The signing CA may ignore or change these values.
Add SAN Row	+ Add SAN Row
Save	

Authorities	Certificates	Revocation
Search		
Search term		
Both		
Search Clear		
Enter a search string or *nix regular expression to search certificate names and distinguished names.		
Certificates		
Name	Issuer	Distinguished Name
GUI default (68d557ecc66a2)	self-signed	O=pfSense GUI default Self-Signed Certificate, CN=pfSense-68d557ecc66a2
Server Certificate		Valid From: Thu, 25 Sep 2025 16:55:41 +0200
CA: No		Valid Until: Wed, 28 Oct 2026 15:55:41 +0100
Server: Yes		
VPN-SSL-REMOTE-ACCESS	CA-HPK11-OPENVPN	CN=hpk11-serveur-vpn, C=FR
Server Certificate		Valid From: Thu, 27 Nov 2025 17:20:15 +0100
CA: No		Valid Until: Sun, 25 Nov 2035 17:20:15 +0100
Server: Yes		
Add/Sign		

Création des utilisateurs locaux

Dans System → User Manager → Users :

User Properties

Defined by

USER

Disabled

☐ This user cannot login

Username

test-vpn

Password

Enter a new password.

Hints:
Current NIST guidelines prioritize password length over complexity.
The password cannot be identical to the username.

Type the new password again for confirmation.

Full name

Compte VPN

User's full name, for administrative information only

Expiration date

Leave blank if the account shouldn't expire, otherwise enter the expiration date as MM/DD/YYYY

Custom Settings

☐ Use individual customized GUI options and dashboard layout for this user.

Group membership

Observateurs PFSense
admins

Not member of

Member of

Move to "Member of" list

Move to "Not member of" list

Certificate

☒ Click to create a user certificate

Create Certificate for User

Descriptive name

VPN-SSL-USER

Certificate authority

CA-HPK11-OPENVPN

Key type

RSA

2048

The length to use when generating a new RSA key, in bits.
The Key Length should not be lower than 2048 or some platforms may consider the certificate invalid.

Digest Algorithm

sha256

The digest method used when the certificate is signed.
The best practice is to use an algorithm stronger than SHA1. Some platforms may consider weaker digest algorithms invalid

Lifetime

3650

Configurer le serveur OpenVPN

Créer un nouveau serveur :

VPN / OpenVPN / Servers

Servers

Clients

Client Specific Overrides

Wizards

OpenVPN Servers

Interface	Protocol / Port	Tunnel Network	Mode / Crypto	Description	Actions
+ Add					

General Information

Description

A description of this VPN for administrative reference.

Disabled

☐ Disable this server

Set this option to disable this server without removing it from the list.

Mode Configuration

Server mode

Remote Access (SSL/TLS + User Auth)

Backend for authentication

Liaison pfSense AD

Local Database

Device mode

tun - Layer 3 Tunnel Mode

"tun" mode carries IPv4 and IPv6 (OSI layer 3) and is the most common and compatible mode across all platforms.

"tap" mode is capable of carrying 802.3 (OSI Layer 2.)

Endpoint Configuration

Protocol

UDP on IPv4 only

Interface

WAN

The interface or Virtual IP address where OpenVPN will receive client connections.

Local port

1194

The port used by OpenVPN to receive client connections.

Cryptographic Settings

TLS Configuration

☒ Use a TLS Key

A TLS key enhances security of an OpenVPN connection by requiring both parties to have a common key before a peer can perform a TLS handshake. This layer of HMAC authentication allows control channel packets without the proper key to be dropped, protecting the peers from attack or unauthorized connections.The TLS Key does not have any effect on tunnel data.

☒ Automatically generate a TLS Key.

Peer Certificate Authority

CA-HPK11-OPENVPN

Peer Certificate Revocation list

No Certificate Revocation Lists defined. One may be created here: [System > Cert. Manager](#)

OCSP Check

☐ Check client certificates with OCSP

Server certificate

VPN-SSL-REMOTE-ACCESS (Server: Yes, CA: CA-HPK11-OPENVPN)

Certificates known to be incompatible with use for OpenVPN are not included in this list, such as certificates using incompatible ECDSA curves or weak digest algorithms.

DH Parameter Length

2048 bit

Diffie-Hellman (DH) parameter set used for key exchange.

ECDH Curve

Use Default

The Elliptic Curve to use for key exchange.

The curve from the server certificate is used by default when the server uses an ECDSA certificate. Otherwise, secp384r1 is used as a fallback.

Data Encryption Algorithms

AES-128-CBC (128 bit key, 128 bit block)

AES-128-CFB (128 bit key, 128 bit block)

AES-128-CFB1 (128 bit key, 128 bit block)

AES-128-CFB8 (128 bit key, 128 bit block)

AES-128-GCM (128 bit key, 128 bit block)

AES-128-OFB (128 bit key, 128 bit block)

AES-192-CBC (192 bit key, 128 bit block)

AES-192-CFB (192 bit key, 128 bit block)

AES-192-CFB1 (192 bit key, 128 bit block)

AES-192-CFB8 (192 bit key, 128 bit block)

Available Data Encryption Algorithms

Click to add or remove an algorithm from the list

The order of the selected Data Encryption Algorithms is respected by OpenVPN. This list is ignored in Shared Key mode.

AES-128-GCM

Allowed Data Encryption Algorithms. Click an algorithm name to remove it from the list

Tunnel Settings

IPv4 Tunnel Network

10.10.10.0/24

This is the IPv4 virtual network or network type alias with a single entry used for private communications between this server and client hosts expressed using CIDR notation (e.g. 10.0.8.0/24). The first usable address in the network will be assigned to the server virtual interface. The remaining usable addresses will be assigned to connecting clients.

A tunnel network of /30 or smaller puts OpenVPN into a special peer-to-peer mode which cannot push settings to clients. This mode is not compatible with several options, including Exit Notify, and Inactive.

IPv6 Tunnel Network

This is the IPv6 virtual network or network type alias with a single entry used for private communications between this server and client hosts expressed using CIDR notation (e.g. fe80::/64). The ::1 address in the network will be assigned to the server virtual interface. The remaining addresses will be assigned to connecting clients.

Redirect IPv4 Gateway

☐ Force all client-generated IPv4 traffic through the tunnel.

Redirect IPv6 Gateway

☐ Force all client-generated IPv6 traffic through the tunnel.

IPv4 Local network(s)

192.168.111.0/24, 192.168.112.0/24, 192.168.113.0/24, 192.168.114.0/24

IPv4 networks that will be accessible from the remote endpoint. Expressed as a comma-separated list of one or more CIDR ranges or host/network type aliases. This may be left blank if not adding a route to the local network through this tunnel on the remote machine. This is generally set to the LAN network.

IPv6 Local network(s)

IPv6 networks that will be accessible from the remote endpoint. Expressed as a comma-separated list of one or more IP/PREFIX or host/network type aliases. This may be left blank if not adding a route to the local network through this tunnel on the remote machine. This is generally set to the LAN network.

Concurrent connections

10

Specify the maximum number of clients allowed to concurrently connect to this server.

Activer Windows

Accédez aux paramètres pour activer Windows Defender.

Client Settings

Dynamic IP

☒ Allow connected clients to retain their connections if their IP address changes.

Topology

net30 - Isolated /30 network per client

Specifies the method used to supply a virtual adapter IP address to clients when using TUN mode on IPv4. Some clients may require this be set to "subnet" even for IPv6, such as OpenVPN Connect (iOS/Android). Older versions of OpenVPN (before 2.0.9) or clients such as Yealink phones may require "net30".

Advanced Client Settings

DNS Default Domain

☒ Provide a default domain name to clients

DNS Default Domain

hpk11.labo

DNS Server enable

☒ Provide a DNS server list to clients. Addresses may be IPv4 or IPv6.

DNS Server 1

172.16.111.1

DNS Server 2

DNS Server 3

DNS Server 4

Block Outside DNS

☐ Make Windows 10 Clients Block access to DNS servers except across OpenVPN while connected, forcing clients to use only VPN DNS servers. Requires Windows 10 and OpenVPN 2.3.9 or later. Only Windows 10 is prone to DNS leakage in this way, other clients will ignore the option as they are not affected.

Force DNS cache update

☐ Run "net stop dnscache", "net start dnscache", "ipconfig /flushdns" and "ipconfig /registerdns" on connection initiation. This is known to kick Windows into recognizing pushed DNS servers.

NTP Server enable

☐ Provide an NTP server list to clients

NetBIOS enable

☐ Enable NetBIOS over TCP/IP

If this option is not set, all NetBIOS-over-TCP/IP options (including WINS) will be disabled.

Activer Windows

Accédez aux paramètres pour activer Windows Defender.

Advanced Configuration

Custom options

auth-nocache

Enter any additional options to add to the OpenVPN server configuration here, separated by semicolon. EXAMPLE: push "route 10.0.0.0 255.255.255.0"

Username as Common

☐ Use the authenticated client username instead of the certificate common name (CN).

Save et configuration VPN faite !

VPN / OpenVPN / Servers

Servers Clients Client Specific Overrides Wizards

OpenVPN Servers

Interface	Protocol / Port	Tunnel Network	Mode / Crypto	Description	Actions
WAN	UDP4 / 1194 (TUN)	10.10.10.0/24	Mode: Remote Access (SSL/TLS + User Auth) Data Ciphers: AES-128-GCM, AES-256-CBC Digest: SHA256 D-H Params: 2048 bits		

[+ Add](#)

Exporter la configuration OpenVPN

Installer un paquet pour l'exportation :

System / Package Manager / Available Packages

Installed Packages Available Packages

Search

Search term: Both [Search](#) [Clear](#)

Enter a search string or *nix regular expression to search package names and descriptions.

Packages

Name	Version	Description	
openvpn-client-export	1.9.6	Exports pre-configured OpenVPN Client configurations directly from pfSense software.	+ Install
Package Dependencies: openvpn-client-export-2.6.17 openvpn-2.6.14 zip-3.0_3 7-zip-24.09			
WireGuard	0.2.9_5	WireGuard(R) is an extremely simple yet fast and modern VPN that utilizes state-of-the-art cryptography. It aims to be faster, simpler, leaner, and more useful than IPSec, while avoiding the massive headache. It intends to be considerably more performant than OpenVPN. WireGuard is designed as a general purpose VPN for running on embedded interfaces and super computers alike, fit for many different circumstances. Initially released for the Linux kernel, it is now cross-platform and widely deployable. It is currently under heavy development, but already it might be regarded as the most secure, easiest to use, and simplest VPN solution in the industry.	+ Install

Dans OpenVPN → Client Export Utility → Client Export :

Advanced

Additional configuration options

auth-nocache

Enter any additional options to add to the OpenVPN client export configuration here, separated by a line break or semicolon.

EXAMPLE: remote-random;

Ensuite on peut télécharger la config en sélectionnant 'Most Clients' :

Save as default

Search

Search term

Search

Clear

Enter a search string or *nix regular expression to search.

OpenVPN Clients

User	Certificate Name	Export
Certificate with External Auth	VPN-SSL-USER	- Inline Configurations: Most Clients Android OpenVPN Connect (iOS/Android) - Bundled Configurations: Archive Config File Only - Current Windows Installer (2.6.17-lx001): 64 bit 32 bit - Legacy Windows Installers (2.4.12-lx601): 10/2016/2019 7/8/8.1/2012r2 - Viscosity (Mac OS X and Windows): Viscosity Bundle Viscosity Inline Config

Créer les règles de firewall pour OpenVPN

Autoriser le flux OpenVPN

Dans Firewall → Rules → WAN, créer une nouvelle règle :

Firewall / Rules / Edit

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

WAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

UDP

Choose which IP protocol this rule should match.

Activier Windows
Accédez aux paramètres pour activer Wind

Destination

Destination

☐ Invert match

WAN address

Destination Address

/

Destination Port Range

OpenVPN (1194)

From

Custom

To

OpenVPN (1194)

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☒ Log packets that are handled by this rule
Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

⚙ Display Advanced

Save

Valider et appliquer la règle

Autoriser les flux vers les ressources

Ajouter une nouvelle règle dans section OpenVPN :

Firewall / Rules / Edit

⚙ ⌨ ⌨ ?

Edit Firewall Rule

Action

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule
Set this option to disable this rule without removing it from the list.

Interface

Choose the interface from which packets must come to match this rule.

Address Family

Select the Internet Protocol version this rule applies to.

Protocol

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

Source Address

/

⚙ Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, **any**.

Destination

Destination

☐ Invert match

172.29.6.13

/

Destination Port Range

(other)

3389

(other)

3389

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☒ Log packets that are handled by this rule
Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

⚙ Display Advanced

Activer Windows

Accédez aux paramètres pour activer Wind

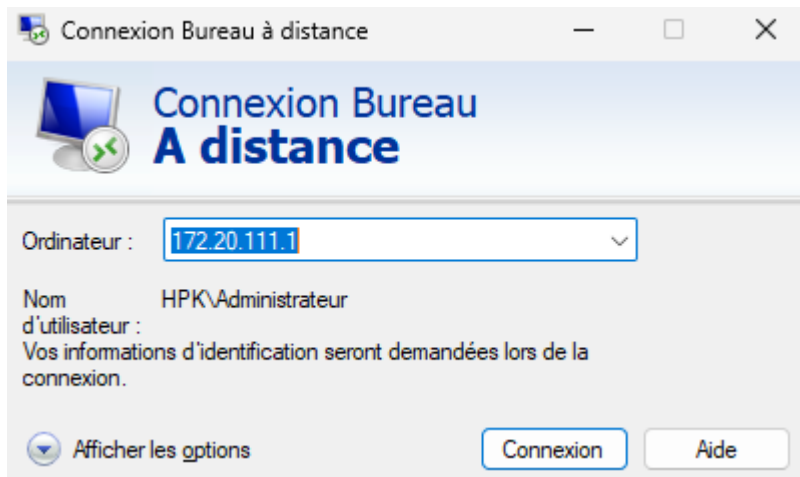
Mettre l'ip du poste

Test de l'accès en VPN depuis le poste

Installer :

[Open Source Community](#) | [OpenVPN](#)

Une fois l'installation faite, lancer un bureau à distance :



Copier le fichier téléchargé dans Documents du PC et le mettre ensuite dans
C:/Programmes/OpenVPN/config

Appuyer sur 'Se connecter' et se connecter avec l'utilisateur pfsense test (pfsense et mdp Admin01) :

